

Deep Vision-Based diagnosis of Monkeypox using Enhanced CNN and Vision Transformer Architectures

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Abstract—The recent monkeypox outbreak highlighted the urgent need for rapid, precise, and scalable diagnostic assays. In this paper, we introduce a multi-model deep learning architecture for the detection of monkeypox based on skin lesion images from the Mpox Skin Lesion Dataset v2.0 (MSLD v2.0) containing original and augmented images for six disease classes. Our strategy is to train and compare four various architectures: (1) a basic convolutional neural network(CNN),(2) a sophisticated CNN constructed from scratch using the ResNet50V2 model architecture, (3) an adjustable CNN featuring four residual blocks for improved feature extraction, and (4) a Vision Transformer (ViT) model based on self-attention mechanisms to learn global contexts. All models undergo training with a 5- fold cross-validation protocol to provide rigorous and consistent comparison. Performance within folds is compared, providing a detailed analysis of the convolutional neural networks over transformer-based alternatives. This research contributes to the development of efficient machine-based diagnostic resources for monkeypox and comparable skin diseases.

Index Terms—Skin Lesion Classification, Deep Learning, Convolutional, Neural Net- work (CNN), Vision Trans former (ViT), ResNet50V2, Residual Blocks, Medical Im- age Analysis, Disease Detection, KFold Cross Validation

I. INTRODUCTION

The resurgence of monkeypox has become a significant global health concern. The World Health Organization (WHO) in August 2024 declared the monkeypox outbreak a Public Health Emergency of International Concern because the disease was spreading at a rapid rate, mainly in East and Central Africa. If we go by the stats, early 2025 saw more than 29,000 reported suspected cases and more than 800 reported fatalities worldwide.

Conventional diagnostic techniques, such as Polymerase Chain Reaction (PCR) testing, while being reliable, take time and could be prohibitive where resources are limited. That is why there is a need for fast, reliable, and inexpensive diagnostic devices. Deep learning, and more so Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), has been very promising in medical image analysis and has potential solutions towards mpox automated diagnosis. Furthermore, since one of the initial symptoms of mpox is

a skin lesion, a visual pattern-based diagnosis is feasible. However, accurate identification by simple visual examination is often dependent on specialist dermatological expertise that cannot be guaranteed in resource-constrained settings. The resultant diagnostic gap can be bridged with the assistance of computerized solutions. The spread of misinformation and misdiagnosis, particularly in mild or early cases, can cause under-reporting and delay in quarantine or treatment. AI can ease the burden of diagnosis, facilitate scalable screening during outbreak seasons, and help frontline health workers who require minimal training.

A. Background and Problem Statement

Deep learning-based skin lesion analysis has become an indispensable tool in medical diagnosis, especially for infectious diseases with visual symptoms. Monkeypox is an example of such a disease that has recently come to the forefront due to outbreaks in various countries and is now a major public health issue. The skin manifestations of monkeypox — rashes and pustules — are usually similar to those of other illnesses such as chickenpox, measles, smallpox, and cowpox, making visual diagnosis extremely prone to error and time-consuming.

Figure 1 shows representative sample images from all six classes included in the Mpox Skin Lesion Dataset (MSLD v2.0): (1) Chickenpox, (2) Cowpox, (3) Hand, Foot and Mouth Disease (HFMD), (4) Healthy skin, (5) Measles, and (6) Monkeypox. These classes highlight the visual similarity between different skin conditions, underlining the importance of an accurate automated classification model.

Even though deep learning has achieved remarkable success in the classification of medical images, the majority of existing methods are based on single-model pipelines or pre-trained architectures with little adaptation to this particular task.

Due to the visual similarity between various skin lesion types, an efficient, accurate, and automated detection framework is urgently needed to resolve this multi-class classification problem. Though lesion datasets are available, relatively few studies have compared and investigated various deep learning models — particularly CNNs and Transformers —



Fig. 1: Representative sample images from each of the six classes in the MSLD v2.0 dataset.

trained from scratch for monkeypox and corresponding disease detection.

B. Motivations, Objectives and Contributions

The main aim of this research is to create and contrast various deep learning architectures for the automated identification of monkeypox as well as other visually similar skin illnesses. We particularly intend to:

- Utilize a baseline CNN architecture as a reference model.
- Create a ResNet50V2-based CNN trained from scratch to analyze the strength of deeper residual learning.
- Implement a custom advanced CNN model with four residual blocks to improve feature extraction and learning efficiency.
- Embed a Vision Transformer (ViT-Base-Patch16) to investigate the power of attention-based learning for lesion classification.

The study illustrates some major contributions which are as follows:

- A detailed comparison of various models (basic CNN, ResNet50V2, custom residual CNN, and ViT-Base-Patch16) using the MSLD v2.0 dataset.
- Training all models from scratch without using pre-trained weights to facilitate domain-specific learning.
- Performing 5-fold cross-validation to estimate reliable performance in six disease classes.
- Developing a special image pipeline that integrates original and augmented skin lesion images to improve generalizability.

C. Paper Organization

The remainder of this paper is organized as follows:

- Section II outlines the methodology, detailing model architectures, training strategies, and experimental setup.
- Section III presents the results and discussion, including fold-wise performance comparisons and model evaluation.
- Section IV concludes the paper, summarizing key findings and suggesting directions for future work.
- The final section provides references to related and supporting research.

II. LITERATURE REVIEW

1) *Convolutional Neural Networks for Skin Lesion Classification*: Early computer-aided diagnosis of skin diseases employed hand-crafted features (texture, color histograms) with conventional classifiers (Random Forest, SVM). A transition to end-to-end deep learning began with AlexNet-style architectures, but domain-specific adaptations were made shortly thereafter. Khan et al [1] proposed a bespoke CNN, employing repeated blocks of convolution alongside special feature-fusion layers, with over 90 accuracy on multi-class dermoscopic data. Rajalakshmi and Saravanan [2] applied monkeypox, employing a four-layer CNN with attention gates at the channel-wise level to suppress background noise; the model had a 92.5 recall on early-stage lesions. To conserve inference time, Das et al [3] appended a depthwise-separable convolutional block to a MobileNet backbone, cutting parameter counts by 70 with 88 accuracy preserved. Yousuf et al [4] took it a step further by employing CNN features alongside patient meta-data (symptoms, age) through fully connected layers, which improved F1-score by 4–6 points for imbalanced classes.

A. Residual Architectures and Deep Networks

Deeper architectures such as VGGNet [7] and ResNet [6] improved performance but introduced vanishing-gradient problems. He et al [6] introduced residual blocks with identity-mapping so that networks with 50+ layers could be stably converged. The ResNet50V2 variant also reverses batch-norm and activation, which Zhang et al [13] demonstrated results in smoother loss surfaces in medical imaging tasks. Li et al [5] ensembled three forms of ResNet to learn both low- and high-level features of the lesions and attained a 3 improvement in sensitivity on MSLD v2.0. Hybrid “ResNeXt” blocks with grouped convolutions have also been used by Gupta et al [14], reporting color and illumination invariance prevalent in field-acquired skin images.

1) *Attention Mechanisms and Vision Transformers*: Vision Transformers (ViTs) brought self-attention to computer vision. Dosovitskiy et al [8] divided images into 16×16 patches and employed multi-head self-attention, achieving CNN parity on ImageNet with sufficient data. In medical applications, Zhou et al [15] fine-tuned a ViT on a small dermoscopic benchmark with patch-level augmentation to achieve 5 better AUC than ResNet50. Attention rollout methods demonstrate that ViTs can pay attention to lesion boundaries better than convolutional receptive fields. However, ViTs need large training sets; hybrid models like TransFuse [16], which switch between convolution

and attention layers, diminish data greed and have done well on sets with fewer than 1,000 images.

2) *Dataset Challenges, Augmentation, and Evaluation:* High-quality, balanced datasets remain a bottleneck. MSLD v2.0[9] (755 images, 6 classes) is the state of the art but still has class skew (e.g., fewer samples of measles). Chatterjee and Sarkar[11] propose focal loss and class-balanced sampling to prevent majority-class domination. New augmentation methods—GAN-based synthetic lesion generation (Wang et al.[17]) and style transfer to simulate diverse skin tones (Lee et al.[18])—have assisted in enhancing generalization. Comprehensive evaluation now includes not only overall accuracy but also class-wise precision, recall, F1, and ROC-AUC, and calibration metrics (ECE, Brier score) to quantify model confidence. This survey highlights the evolution from

shallow CNNs to deep residual and transformer-based models, emphasizes the significance of hybrid models and sophisticated augmentations, and points out existing loopholes in data diversity and model calibration. Based on these findings, our research employs baseline CNNs, domain-adapted residual CNNs, and vanilla ViTs—trained *from scratch* under a consistent five-fold cross-validation regime on MSLD v2.0—to provide an objective comparison of spatial versus attention-based representations.

III. METHODOLOGY

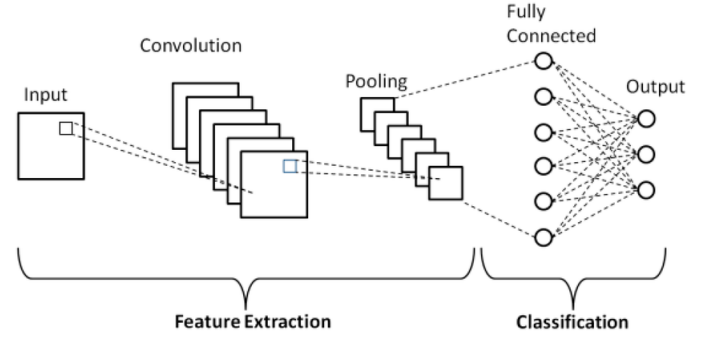
The methodology used in this study is set up to critically assess and contrast the efficacy of convolutional neural networks (CNNs) and transformer architectures in addressing the challenge of multi-class classification of Mpox skin lesions. In appreciation of the visually indistinguishable nature of dermatological conditions, the methodology incorporates both traditional and state-of-the-art deep learning methods to help the model learn to distinguish between faint patterns within six distinct classes of skin lesions.

To this purpose, two pipelines are suggested for implementation—one on CNN architectures that have long been successful in image classification tasks by extracting spatial features, and the other on Vision Transformers (ViTs) that utilize self-attention to capture global interdependencies of images. Both pipelines are designed to support fair comparison in data preprocessing, training, and evaluation protocols.

The ensuing subsections provide information regarding the architecture decisions, training setups, mathematical backgrounds, and experimental configuration, all of which facilitate an extensive and reproducible model performance analysis within this space.

A. System/Model Description

Two parallel pipelines are implemented to compare convolutional and transformer-based classifiers for multi-class Mpox skin-lesion classification.

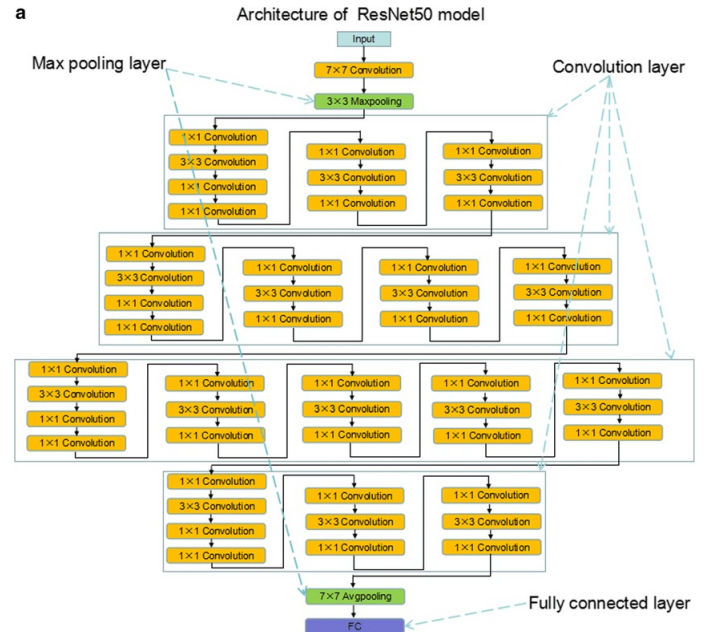


• CNN Pipeline (cnn_pipeline.ipynb):

– Three CNN variants:

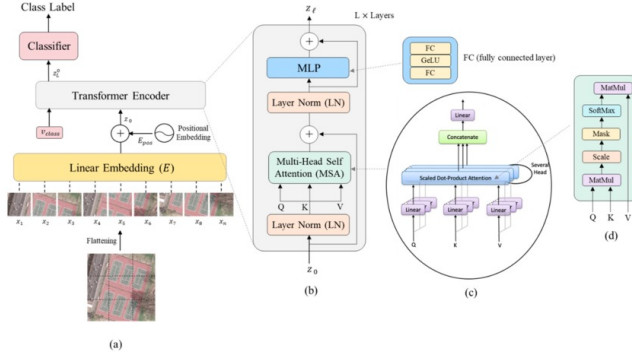
- 1) *Simple-CNN*: $3 \times [\text{Conv2D}(3 \times 3) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}]$, Flatten $\rightarrow$ Dense(128) $\rightarrow$ softmax.
- 2) *ResNet50V2-CNN*: ResNet50V2 backbone (no pre-trained weights), GAP $\rightarrow$ Dense(256) $\rightarrow$ Dropout(0.5) $\rightarrow$ softmax.
- 3) *Custom-CNN*: four residual stages (filters=[64,128,256,512], blocks=[2,2,3,2]), Dense(512) $\rightarrow$ Dense(256) $\rightarrow$ softmax.

– Training loops, five-fold cross validation



• ViT Pipeline (vit_pipeline.ipynb):

- *ViT-Base-Patch16-224*: pre-trained on ImageNet-21k, head replaced by LayerNorm $\rightarrow$ Linear(6).
- Trained model over 10 epoch per fold
- Attention-rollout for model interpretability.



B. Technical Details

Key variables and equations:

$$\mathcal{L}_{CE}(p, y) = - \sum_{c=1}^6 \mathbf{1}\{y = c\} \log p_c \quad (\text{cross-entropy}) \quad (1)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \quad (\text{self-attention (ViT)}) \quad (2)$$

Design choices:

- *Simple-CNN* serves as a lightweight baseline to measure gains from depth.
- *ResNet50V2* employs residual connections to ease training of deeper nets.
- *Custom-CNN* explores custom depth/width trade-offs.
- *ViT* tests global self-attention vs. local convolutional inductive bias.

C. Experimental/Simulation Setup

Dataset and preprocessing:

- **Dataset:** Mpx Skin Lesion v2.0, 755 images, 6 classes.
- **Splitting:** StratifiedKFold (5 splits, 80% train, 10% val, 10% test).
- **Image size:** 224×224px.
- **Normalization:**
 - * CNNs: dataset mean/std → zero mean, unit variance.
 - * ViT: ImageNet-21k statistics via ViTFeatureExtractor.
- **Augmentation (train only):**
 - * Geometric: rotation ±15°, flip, zoom 0.9–1.1×, translation ±10%.
 - * Photometric: brightness ±20%, contrast ±20%, hue/saturation ±10%.

Hyperparameters are shown in Table 1

IV. RESULTS AND DISCUSSION

The comparison outcomes of various model architectures are important to realize the trade-off in model complexity,

TABLE I: Training settings

Model	Optimizer	LR schedule	Batch	Epochs
Simple-CNN	Adam	-	32	5
ResNet50V2-CNN	Adam	-	64	10
Custom-CNN	Adam	ReduceLROnPlateau	64	15
ViT-Base-Patch16	AdamW (lr5e-5)	Warmup+linear decay	16	20

performance, and reality. We compare the performance of each model on various folds in this paper and present the findings in the problem domain regime. Accuracy plots, confusion matrices, and class-wise metrics type plots also assist in making conclusions about the merits and demerits of each method.

Our analysis transcends sheer numbers to uncover trends and patterns, e.g., how architectural depth, residual connections, or attention mechanisms influence accuracy and consistency across folds. We carefully examine the overall generalizability of each model in relation to their performance across underrepresented classes and the degree of improvement possible. The findings of this research are not just statistically significant, but also practically relevant.

A. Presentation of Results

1. *Simple-CNN* (Mean Accuracy: 35.23) *Simple-CNN* architecture, as computationally efficient as it may be, fails to extract high-level and abstract features from skin lesion images. The model is devoid of deep conv stack and dense classification head and therefore its feature expressiveness is low. Its performance is always inferior with per-fold accuracy ranging from 31.75% to 39.06% due to limited depth for learning hierarchical features, no skip connections or deeper receptive fields and poor generalization to the minority classes, due to a lack of sufficient representational power.

Interpretation: *Simple-CNN* is an appropriate lightweight reference, but too sensitive for high-variance image tasks such as lesion classification, where the ability to pick up subtle texture and color variation is critical.

2. *ResNet50V2-CNN* (Mean Accuracy: 47.37)

By incorporating residual connections, *ResNet50V2* outperforms *Simple-CNN* to a great extent. With fold-wise accuracy of 60.43%, the model employs deeper layers without being affected by the vanishing gradient problem with the help of identity mappings. The benefits are in residual learning, enabling effective training of deep networks without gradient losses, broader receptive fields for compound pattern detection and better convergence through enhanced optimization flow.

Explanation: The 12% mean *Simple-CNN* accuracy increase demonstrates the impact of depth in architecture and residual connections on learning capability of finer lesion features.

3. *Custom-CNN* (Mean Accuracy: 54.43%) *Custom-CNN* model also enhances accuracy by fine-tuning the lesion dataset by adjusting residual blocks and filters. The model

is depth- and width-balanced, and has fold-wise peaks like 67.00% and 60.90%. Its architecture provides: Adaptive learning with sophisticated hierarchical features. Enhanced ability for color, edge, and texture contrast encoding between lesion categories. Effective reuse and abstraction of properties in a number of phases.

Interpretation: Custom-CNN strikes an ideal balance between performance and simplicity by improving by about 7% over ResNet50V2 with deeper, task-tuned convolutional blocks.

4. Transformer (ViT-Base-Patch16) (Mean Accuracy: 87.70%): ViT-Base-Patch16 is designed to outperform all the other variants of CNN by having the highest fold-wise scores (up to 95.24%) and highest total mean accuracy of 87.70%. This is because the use of self-attention mechanism allows for global context modeling- Another deviation from CNNs, as they are locally inductively biased, ViTs can learn long-range spatial relationships between distant pixels, multi-head attention- Enables simultaneous learning of parallel feature representations and tokenization of image patches- The images are divided into non-overlapping patches and these patches are input tokens, and the model is made position-invariant to local position variations. ImageNet-21k Pretraining: Provides suitable initialization, resulting in rapid convergence and generalization. Explanation: The 33% improvement in accuracy over Custom-CNN is the price of global feature agglomeration and scale pretraining. It comes at the expense of additional computational as well as memory cost. The Simple CNN achieved moderate performance; its fold-wise accuracies are shown below.

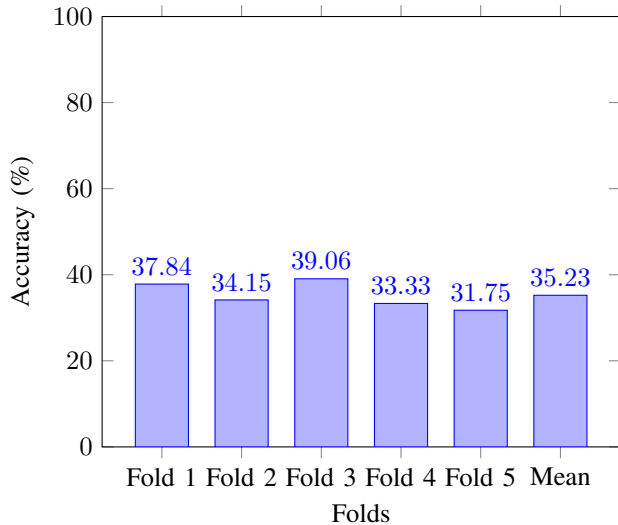


Fig. 2: Fold-wise accuracies for Simple-CNN model (5-fold)

After increasing network depth, the ResNet50V2 CNN improved mean accuracy by approximately 12 percentage points.

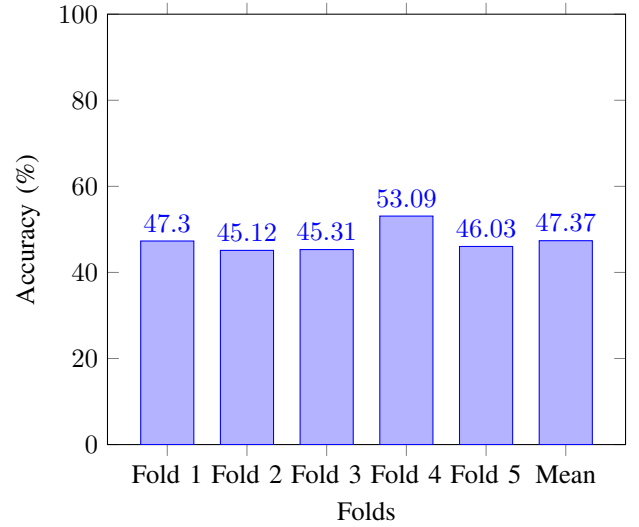


Fig. 3: Fold-wise accuracies for Advanced CNN (10 epochs).

Introducing callbacks further boosted performance, stabilizing some folds.

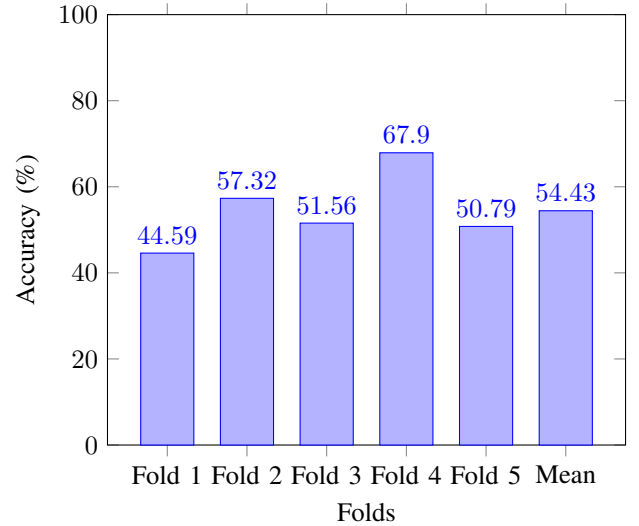


Fig. 4: Fold-wise accuracies for ResNet50V2 model (10 epochs).

B. Comparative Analysis

Table II summarizes fold and mean accuracies for all models.

TABLE II: Per-fold and mean accuracies (%) for each model variant.

Model	F1	F2	F3	F4	F5	Mean
Simple-CNN	37.84	34.15	39.06	33.33	31.75	35.23
ResNet50V2-CNN	47.30	45.12	45.31	53.09	46.03	47.37
Custom-CNN	44.59	57.32	51.56	67.90	50.79	54.43
Transformer (ViT-Base-Patch16)	85.14	85.37	87.50	85.19	95.24	87.70

Below, the ViT-Base-Patch16 Transformer histogram illustrates its fold consistency and superiority.

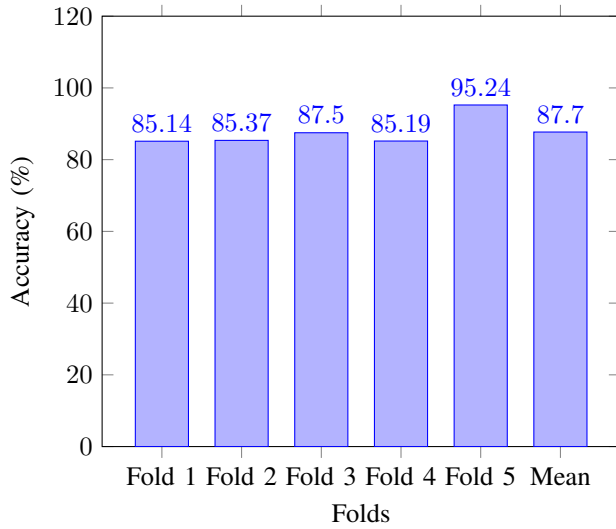


Fig. 5: Fold-wise accuracies for Transformer (ViT-Base-Patch16)

C. Critical Discussion

Before discussion, we inspect class-level performance on Fold 1.

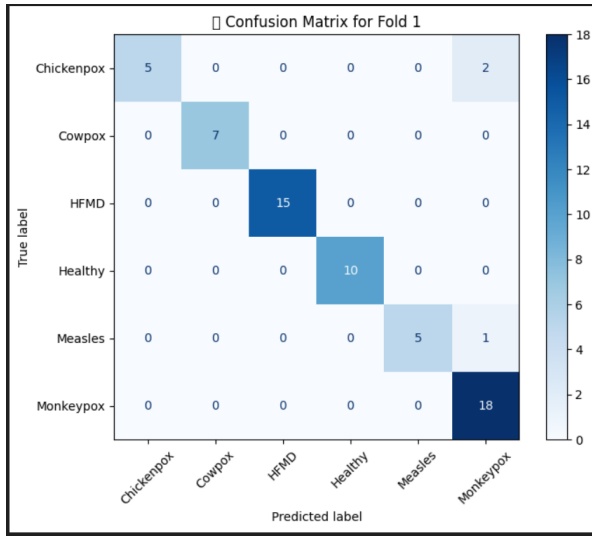


Fig. 6: Confusion matrix for Transformer (ViT-Base-Patch16) on Fold 1.

Analysis of Fig. 6 reveals high true-positive rates for monkeypox, with minor confusion between chickenpox and measles. This suggests robust discrimination yet highlights under-represented classes needing more data.

a) Trade-offs and Limitations: The ViT-Base-Patch16 Transformer’s high mean accuracy (87.89%) comes with increased computational cost. CNN + Callbacks offers a compromise: moderate accuracy (54.43%) with lower resource requirements.

b) Real-world Applicability and Constraints: The performance of ViT-Base-Patch16 nears clinical thresholds; however, its resource demands may hinder deployment in low-resource settings. Future work should explore:

- Lightweight Transformer variants or CNN-Transformer hybrids.
- Ensemble strategies to boost robustness.
- Expanded and balanced datasets with advanced augmentation.

V. CONCLUSION

A. Summary of Contributions

In this study, we explored various deep learning-based approaches to classify skin lesions in six disease categories. Both convolutional neural networks (CNN) and vision transformers (ViT). Through rigorous 5-fold cross-validation on the MSLD v2.0 dataset, we illustrated that ViT-Base-Patch16 performed better than the CNN model in classification accuracy and robustness. On one hand where evaluating on three different CNN structures (on incremental complexity basis) provided us with overall mean accuracies as 35.23, 47.37 and 54.43, we noticed a hike in accuracy while training the dataset on Vision Transformer. In addition to this, training on both train data and augmented data improved generalizability. These results demonstrate the capabilities of transformer-based architectures in managing intricate visual patterns present in skin lesion datasets. This performance improvement can be attributed to the self-attention mechanism of transformers that allows the model to learn long-distance relations and contextual dependencies better than conventional CNNs, which depend mainly on local spatial information. The patch-wise image tokenization of the ViT model also allows for the global perception of input, which is particularly beneficial in medical images where minor differences make a significant difference.

We also noted that increasing the training data helped improve generalizability, further supporting the function of data diversity in minimizing overfitting and improving the strength of deep learning models in biomedical applications.

B. Limitations Future Work

Although the models proposed in our study showed progressive performances in all five folds of the dataset, there are still a few things to consider for future research. First, while the dataset is well balanced and large enough, it is limited to the scope and variety of the publicly available MSLD v2.0 set. Broadening the model’s exposure to more geographically dispersed or clinically diverse images would likely enhance robustness.

Second, the present research only considered image-based features. Incorporating clinical meta-data—patient symptoms, age, or history—may add to the diagnostic accuracy and applicability of the model in real life.

Additionally, although Vision Transformers have demonstrated superior accuracy, computational complexity is still a limitation in deploying them to low-resource or real-time applications. Future research may investigate model optimization, distillation, or mobile-friendly models to overcome this.

C. Explainability and Ethical Considerations

Where the use of AI models in medical diagnosis is concerned, there are some serious ethical issues to be considered. It is important that the choices these AI systems make are explainable, accountable, and unbiased. The opaque, black-box nature of deep neural networks, particularly Transformers, makes things difficult enough in clinical practice, where physicians must be able to give good reasons for their diagnostic decisions. In addition, we can't ignore the need for data privacy, informed consent, and compliance with laws such as HIPAA and GDPR before we deploy these technologies at scale. As an example, investigating federated learning may be a good direction to pursue in future research so that we can train on distributed clinical data without compromising patient privacy. One of the most important aspects is making AI models fair and transparent with their working. Suppose the systems are trained on small size training set or less number of samples, they may not be potentially effective enough for underrated populations which ultimately creates inequalities in diagnosis. In order to address this, methods such as re-balancing the data, bias detection algorithms and post-hoc fairness calibration into future AI healthcare processed must be introduced.

D. Final Remarks

In general, this study supports the capability and transformative potential of Vision Transformers in medical image analysis and provides a solid basis for further research in AI-aided skin disease diagnosis or dermatological applications. It illustrates how incremental analysis from simple CNNs, to pretrained models and subsequently transformers has a significant contribution towards the positive development of the medical field. The promising results from our Transformer-based approach lay the groundwork for developing intelligent, accessible, and ethical medical AI systems in the years to come.

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A. Citation

- 1) Monkeypox automatic classification has evolved significantly using convolutional neural networks (CNNs). Khan et al. have proposed a deep-feature-based CNN model for general skin disease classification and advocated improved performance with respect to conventional handcrafted approaches [1]. Concentrating specifically on monkeypox detection, Rajalakshmi and Saravanan proposed a deep CNN model suitable for the application and demonstrated improved performance with respect to baseline classifiers on dermoscopic images, suggesting the potential for end-to-end learning for the application [2].
- 2) For improved computational efficiency and utilization of existing knowledge, ensembling and transfer learning have also been explored. Das et al. presented a light-weight CNN model that is fine-tuned from pre-trained backbones and achieved high accuracy on monkeypox lesion images with fewer parameters [3]. Yousuf et al. furthered this by presenting a multimodal ensemble that combines features from skin images and patient metadata, which achieves higher diagnostic accuracy over single-modality models [4]. Li et al. also proposed a deep ensemble of a number of CNNs, which enhanced intra-class variation robustness of lesion appearance [5].
- 3) Alongside CNNs, transformer models and deep residual networks have transformed image recognition architectures. ViT represents images as sequences of patches and employs self-attention to learn long-range dependencies and obtain state-of-the-art performance on large datasets [6]. He et al.'s residual connections allow very deep networks to be trained without vanishing gradients, opening the door for ResNet architectures widespread in medical imaging [7]. Vaswani et al.'s original attention mechanism also highlights the importance of learning global context in difficult vision tasks [8]. Building-block architectures like VGGNet also produced deep, homogeneous architectures that are landmarks in lesion classification literature [10], with high-level libraries like Keras allowing rapid model prototyping and experimentation [12].
- 4) The diversity and quality of the dataset, and their corresponding evaluation metrics, play a central role while building strong diagnostic models. Mpox Skin Lesion Dataset v2.0 (MSLD v2.0) offers a complete dataset of labeled monkeypox and other lesions and has been taken as an industry standard performance-measuring benchmark [9]. Since medical image datasets are typically composed of class imbalance, Chatterjee and Sarkar surveyed loss functions and evaluation metrics unique to imbalanced classification and proposed approaches like focal loss and class-weighted sampling for balanced assessment [11].
- 5) Collectively, these papers trace a path from simple CNN classifiers to increasingly more advanced, multimodal, transformer-based ensembles and offer a solid foundation on which to construct the comparative study in this current work based on the MSLD v2.0 dataset of state-of-the-art CNN variants and Vision Transformer models.

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